

SOPHIE LIU

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- PhD in a quantitative field with 6+ years' experience in solving real-world problems, delivering data-driven insights and end-to-end solutions with cross-functional teams in healthcare data science, insurance and business. Great passion for continuous learning.
- Proficient in Python/R/SQL/Java programming. Expertise in machine learning, statistical modeling, A/B testing, pattern recognition, risk assessment and customer segmentation. Strong communication, interpersonal, professional writing and time management skills.

SKILLS

Programming: Python, R, SQL, SAS, Java, MLflow, Shiny, Pandas, NumPy, PyTorch, TensorFlow, Scikit-Learn, SciPy, PySpark, Keras

Stats: Machine Learning, Statistical Modeling, Bayesian, Predictive Modeling, Deep Learning, A/B Testing, Causal Inference, Pattern Recognition, Clustering, Decision Trees, Recommendation Systems, NLP, Hypothesis Testing, Data Mining & Visualization, Anomaly Detection, Optimization

Tools & Platforms: Git, Linux, AWS, GCP, Hive, Spark, Flask, Docker, Tableau

EDUCATION

Penn State University

Ph.D. in Biostatistics

Hershey, PA

08/2019 – Present

Georgetown University

M.S in Biostatistics

Washington DC

08/2015 – 12/2016

SELECTED PROJECTS

Home Credit - Credit Risk Model Stability (Kaggle Competition: **Bronze, top 7%**)

- Developed a model measured against feature stability over time to predict which clients are more likely to default on their loans for the Home Credit Group; Given 465 native features, conducted feature engineering and derived aggregate feature and diff features for modeling; Trained LightGBM and CatBoost model, and improved the robustness of the model by using group k-fold.

E-Commerce Multi-Objective Recommender System (Kaggle Competition: **Silver, top 4%**)

- Developed a Python-based recommender system to increase sales for the German online retailer “OTTO”, predicting e-commerce clicks, cart additions, and orders simultaneously. Leveraged various recall methods for candidate generation with extensive retail data from 12M users, trained and fine-tuned LightGBM for optimal ranking. Achieved a top-tier leaderboard score: top 4%.

SELECTED WORK EXPERIENCE

Penn State University

Research Scientist

Hershey, PA

08/2019 – Present

- **Predictive Modeling:** Implemented an advanced XGBoost model to accurately predict the persistence status of early childhood asthma, leveraging electronic medical record (EMR) dataset with over 600 features. Through EDA, Feature Engineering, rounds of feature selection, SMOTE-ENN and continuous refinement, successfully improved model's performance by 16% elevation in recall. Resulted in an expected 20% decrease in unnecessary hospital visits, leading to elevation of targeted patient retention rate.
- **Fraud Detection:** Employed a classification model to predict the potentially fraudulent providers. Cleaned and analyzed claim data from 138k+ patients; Trained, validated, and tuned multiple models with Random Forest having best performance. Discovered key variables in detecting the behaviors of potentially fraudulent providers by SHAP and further analyzed fraudulent patterns.
- **A/B Testing:** Conducted a 16-week randomized trial with 1:1 enrollment ratio to analyze whether mobile health intervention, “Noom Weight” app, can help patients with NASH lose body weight. Observed that NW significantly decreased body weight compared to standard care by t tests, supporting future large-scale use of mobile-based intervention for patients with NASH.

Alexion, AstraZeneca Rare Disease

Biostatistics Intern

Remote

06/2021 – 08/2021

- **Decision-making:** Deployed an interactive R Shiny dashboard for Go/No-Go decision making in early trial stages based on hypothesis testing under different experiment designs, supporting both Frequentist and Bayesian frameworks. Streamlined internal decisions and eliminated reliance on the commercial software “OKGO”, significantly realizing over 60% cost savings.

Brigham and Women's Hospital

Data Analyst

Boston, MA

09/2017 – 06/2019

- **Causal Inference:** Conducted the propensity score matching to successfully elucidate the causal relationship of how post-diagnosis multimorbidity and lifestyle changes contribute to excess mortality of rheumatoid arthritis (RA), leveraging on Medicare claims data.
- **Classification:** Developed a cutting-edge Classifier for Axial Spondyloarthritis (SpA) Patients, marrying the conventional ICD disease codes with narrative data insights derived via NLP; Employed the unsupervised Multimodal Automated Phenotyping (MAP) method with NLP-augmented features, realizing a notable 60% elevation in the number of identified axial SpA patients with sensitivity amplified by 20%; Reduced dependence on manually generated training labels and the need of domain experts.
- **Mail Campaign & Insights:** Performed EDA and identified insights with 100k+ patients' data. Employed Plotly and Tableau for data visualization of various demographics like age, region, treatment history, etc, refining the segmented mail list and better targeting consumers who're more likely to yield responses to new treatments, coupons, or study recruitment.